

Charge sharpening in circuits with discrete symmetries

1 $\mathbb{Z}_3$ symmetry

1.1 Formulation of the problem

We consider a 1D chain of L qutrits, which evolves under brickwork unitary evolution interspersed with single-site measurements. The evolution is symmetric with respect to the operator $\prod_{j=1}^L \tau_j$, where τ_j is a single-site operator acting on site j as

$$\tau_j = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \quad (1)$$

with $\omega = e^{i\frac{2\pi}{3}}$.

Each time step t involves two layers of unitary operators:

$$\mathbb{U}(t) = \bigotimes_{x \text{ even}} U_{x,x+1}(x,t) \bigotimes_{x' \text{ odd}} U_{x',x'+1}(x',t). \quad (2)$$

Here, each $U_{x,x+1}$ is a two-qutrit unitary that acts on sites x and $x+1$. In the computational basis, $U_{x,x+1}$ decomposes into 3 blocks, each block being of size 3×3 :

$$U_{x,x+1} = U^{(0)} \oplus U^{(1)} \oplus U^{(2)}. \quad (3)$$

The superscript label refers to the total charge modulo 3: $U^{(0)}$ acts on the subspace spanned by $\{|00\rangle, |12\rangle, |21\rangle\}$; $U^{(1)}$ acts on the subspace spanned by $\{|01\rangle, |10\rangle, |22\rangle\}$; and $U^{(2)}$ acts on the subspace spanned by $\{|02\rangle, |20\rangle, |11\rangle\}$.

Each $U^{(j)}$ is *Haar random* (drawn independently for each x and t), so the evolution is random within each charge sector, but conserves the charge modulo 3. Following the two layers of unitary evolution, at each time step t independent single-site projective measurements are applied to the system with a rate p (each site has a probability p to be measured).

We then look at the following charge-sharpening problem: starting from an equal-superposition state $|\Psi_0\rangle = \left[\frac{1}{\sqrt{3}} (|0\rangle + |1\rangle + |2\rangle) \right]^{\otimes L}$, consider the state $|\Psi_f\rangle$ after $t = L$ time steps. Denoting by p_j the probability to measure a total charge j in $|\Psi_f\rangle$ (with $j = 0, 1, 2$), we can ask to what extent p_j is concentrated

at a certain value. This is measured by the entropy $S = -\sum_j p_j \ln p_j$, which is further averaged over different realizations of the circuit. Conversely, assuming that $|\Psi_0\rangle$ is in a definite charge sector j , S measures how likely it is to correctly learn j from the results of the measurements performed throughout the evolution (learnability increases with decreasing S).

1.2 Mapping to a classical evolution

The probability p_j to measure a total charge j in $|\Psi_f\rangle$ given a certain realization of the circuit – represented by a specific record of unitaries and measurements (u, m) – can be written as

$$\begin{aligned} p_j &= \|\mathbb{P}_j C(u, m) |\Psi_0\rangle\|^2 = \langle\Psi_0| C^\dagger(u, m) \mathbb{P}_j C(u, m) |\Psi_0\rangle \\ &= \sum_{\vec{k} \in \mathbb{Z}_3^{\otimes L}} \langle\vec{k}| \otimes \langle\vec{k}| \mathbb{P}_j^{\otimes 2} [C(u, m) \otimes C^*(u, m)] |\Psi_0\rangle \otimes |\Psi_0\rangle, \end{aligned} \quad (4)$$

where $\mathbb{P}_j$ is the projector onto the sector with total charge j , and $|\vec{k}\rangle$ are the computational basis states.

We now average over the unitaries comprising the circuit. As they are independent, we can perform the average for each two-qutrit gate individually. We write each such gate as

$$U_{x,x+1} = U^{(0)} P_0 + U^{(1)} P_1 + U^{(2)} P_2, \quad (5)$$

where P_j projects a two-qutrit state onto the subspace with total charge j . Since the $U^{(j)}$ are Haar-random and independent, we get

$$\mathbb{E}_{\{U^{(j)}\} \sim \text{Haar}} [U_{x,x+1} \otimes U_{x,x+1}^*] = \sum_j |I_j\rangle \langle I_j| P_j \otimes P_j, \quad (6)$$

where $|I_j\rangle$ is the maximally entangled state between the two replicas of the subspace with charge j . Within the subspace of replicated two-qutrit states $|k_1 k_2\rangle^{\otimes 2}$ with $k_1 + k_2 = j \pmod 3$, the projector $|I_j\rangle \langle I_j|$ assumes the form

$$|I_j\rangle \langle I_j| = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}. \quad (7)$$

The evolution thus reduces to the evolution of probability vectors, where each two-qutrit unitary can be replaced by a transfer matrix of the form

$$U_{x,x+1} \longrightarrow \sum_j \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} P_j \quad (8)$$